

Standard Model Particles from $GF(27)^*$

All Quantum Numbers Algebraically Forced (Docs. 346 / 347 / 348 / 349)

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Status markers

- [B] *Proved* — algebraically established.
- [K] *Confirmed* — numerically verified.
- [S] *Speculative* — open.
- [X] *Experimental* — established; FFGFT derivation open.

Structure of the diagram

Each fermion is displayed as a concentric ring diagram (Doc. 356). Rings from inside out encode physical properties in order of algebraic depth:

White inner circle	Particle symbol, charge Q , hypercharge Y
Colour ring (quarks only)	$r/g/b$ sectors = colour charge [B] (Doc. 346); innermost after white because gluons confine colour (Confinement)
Middle ring	Left violet = S_d = left-handed [B] ; right orange = S_u = right-handed [B] (Doc. 349); grey = ν_R absent in SM [B]/[S]
Outer ring	Galois orbit colour: navy = O1/O3 Gen. 1/3 leptons; sky blue = O3 Gen. 2 leptons; teal = O2 Gen. 1/3 quarks; berry = O4 Gen. 2 quarks [B] (Doc. 348)
Red dashed box	$SU(2)_L$ doublet pair [B] (Doc. 349)
Black dashed circle	Generation 3

Core result

The fermion quantum numbers of the Standard Model (Q , Y , colour charge, chirality, generation structure) follow algebraically from $GF(27)^*$ without free parameters **[B]** (Docs. 346, 347, 348, 349); the hypercharge $Y = -1$ of the charged leptons remains open (Section 1).

1 Fermions

Leptons — neutrinos QR, charged leptons NQR

Neutrinos (ν_e, ν_μ, ν_τ): QR ($O_1, O_3, Q = 0$); charged leptons (e, μ, τ): NQR ($O_4 = \{7, 8, 11\}, Q = -1$) (Doc. 346 Theorem A **[B]**). Hypercharge $Y = -1$ of the charged leptons does not follow from Doc. 347 Theorem A (there $NQR \rightarrow Y = +1/3$); its derivation is open. All three generations correspond to the three Frobenius cycles of the root orbits of order 3 in $GF(27)^*$ **[B]** (Doc. 348 Theorem A).

Quarks — Non-Quadratic Residues (NQR)

Up-type quarks (u, c, t) and down-type quarks (d, s, b) originate in the NQR sector of $GF(27)^*$. Electric charges $Q \in \{+2/3, -1/3\}$ follow from the NQR bit and $N \bmod 3$ without free parameters **[B]** (Doc. 346). Colour charge $r/g/b$ is forced by the Z_3 charge operator $N \bmod 3$ in $GF(3)$: three colours = three group elements **[B]** (Doc. 346 Theorem C).

2 Gauge Bosons

The 8 gluons arise from $4 \times Z_{13}$ -orbits $\times 2 Z_2$ -parities in $GF(27)^*$ **[B]** (Doc. 339) — the two groups of four are shown in different shades. Electroweak symmetry breaking $SU(2)_L \times U(1)_Y \rightarrow U(1)_{EM}$ is algebraically forced by three Galois building blocks **[B]** (Doc. 355). In FFGFT, $\tilde{T} \cdot m = 1$ replaces the Higgs mechanism (Doc. 019); the boson is experimentally established **[X]**.

3 Open Points

- **Hypercharge of the charged leptons:** $Y = -1$ does not follow from Doc. 347 Theorem A ($NQR \rightarrow Y = +1/3$); open **[S]**.
- **Generation assignment (ordering):** Which orbit element corresponds to which generation (1/2/3)? Not derived **[S]** (Doc. 346).
- **CKM/PMNS angles:** Structure established (Doc. 348 Theorems B,C **[B]**); numerical values require further work (Doc. 348 Theorem D **[S]**).
- **Higgs derivation:** Mechanism established experimentally **[X]**; formal FFGFT elaboration open (Doc. 019).